Year 12 Maths - Working with data

Similar questions - Histograms and the normal distribution

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The masses, in kg, of 120 parcels handled by a courier are summarised below.

Mass m (kg)	$0 \leqslant m < 2$	$2 \leqslant m < 4$	$4 \leqslant m < 8$	$8 \leqslant m < 14$	$14 \leqslant m < 20$
Frequency	9	21	48	30	12

(a) Calculate the frequency density for each class, so that a histogram could be drawn. **[2]**

(b) Estimate the number of parcels with a mass between 3 kg and 6 kg. **[2]**

(c) Estimate the mean mass of a parcel. **[2]**

(d) Estimate the median mass, and explain why it is lower than the mean. **[3]**

2 The masses of apples from an orchard are modelled by the random variable $X \sim \mathrm{N}(160, 15^2)$, where X is in grams.

(a) Find $\mathrm{P}(145 < X < 175)$. **[1]**

(b) Find the probability that an apple chosen at random has mass greater than 190 g. **[2]**

(c) The heaviest 10% of apples are sold as "premium". Find, to 1 decimal place, the minimum mass of a premium apple. **[2]**

(d) In a random sample of 200 apples, estimate how many have a mass between 150 g and 170 g. **[2]**

(e) A large sample has smallest mass 98 g and largest mass 220 g. Use these values to comment on whether the model is appropriate. **[2]**

Question 1

(a) Frequency densities (frequency ÷ class width): 4.5, 10.5, 12, 5, 2

(b) 3 to 4: $10.5 \times 1 = 10.5$. 4 to 6: $12 \times 2 = 24$. Estimate $= 34.5$, i.e. about 35 parcels

(c) Midpoints $1, 3, 6, 11, 17$: $\dfrac{9+63+288+330+204}{120} = \dfrac{894}{120} = 7.45$ kg

(d) 60th value lies in $4 \leqslant m < 8$: $4 + \dfrac{60-30}{48} \times 4 = 6.5$ kg. The distribution is positively skewed (a long tail of heavy parcels pulls the mean up)

Question 2

(a) These limits are $\mu \pm 1\sigma$, so P $= 0.683$

(b) $z = \dfrac{190-160}{15} = 2$, so $\mathrm{P}(X > 190) = 1 - 0.97725 = 0.0228$

(c) $z = 1.2816$ for the top 10%: $160 + 1.2816 \times 15 = 179.2$ g

(d) $\mathrm{P}(150 < X < 170) = \mathrm{P}(-0.667 < Z < 0.667) = 0.495$. Expected $200 \times 0.495 \approx 99$

(e) 98 g is $\dfrac{98-160}{15} = 4.1$ standard deviations below the mean and 220 g is 4 above, which is very unlikely in a sample from this model (the model suggests almost all values within 3 standard deviations: 115 to 205 g). So the model may underestimate the spread (standard deviation too small)